Exploratory Data Analysis

Dataset: Employee_Demo_Dataset

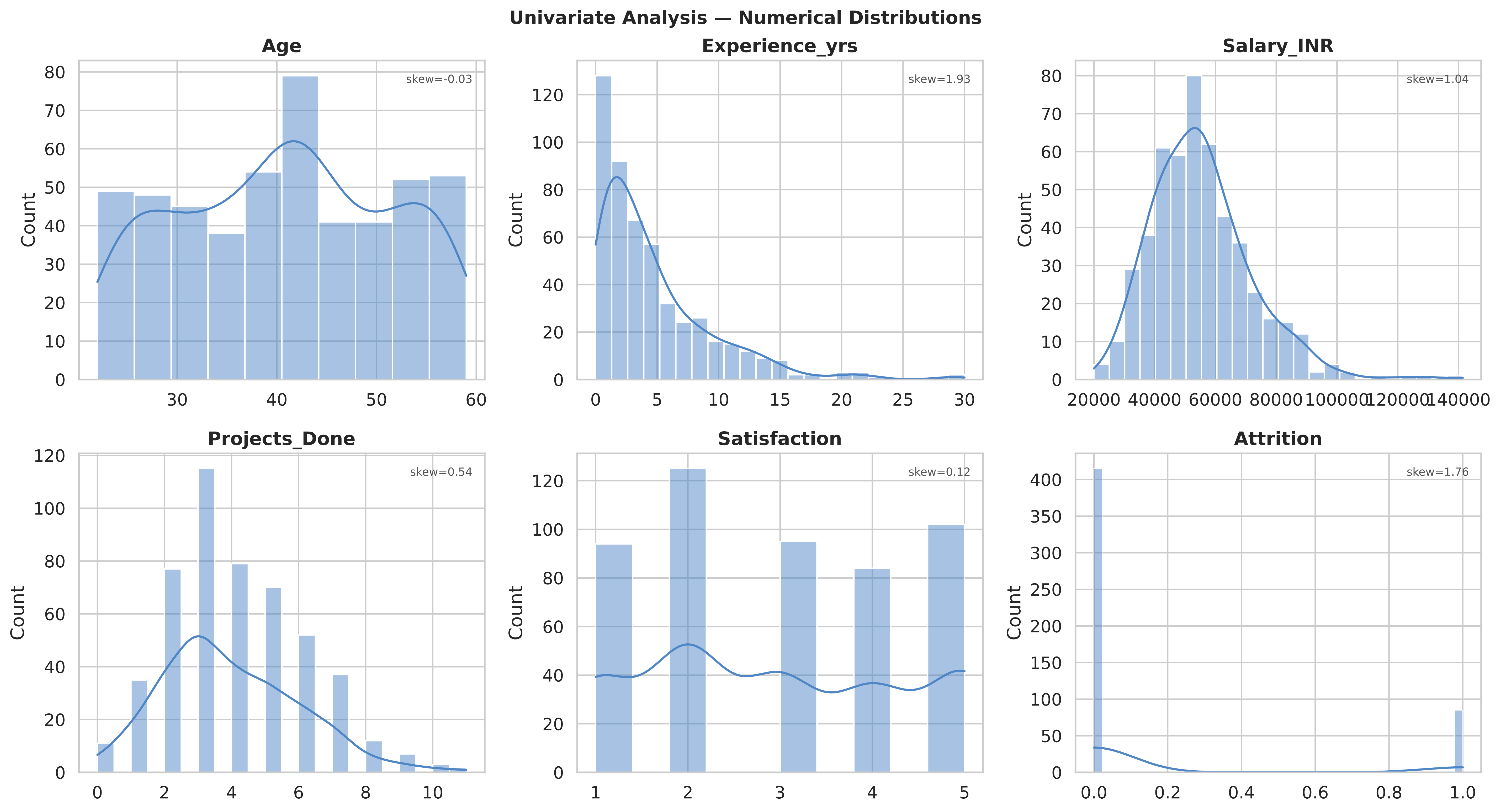
Rows: 515 | Columns: 9

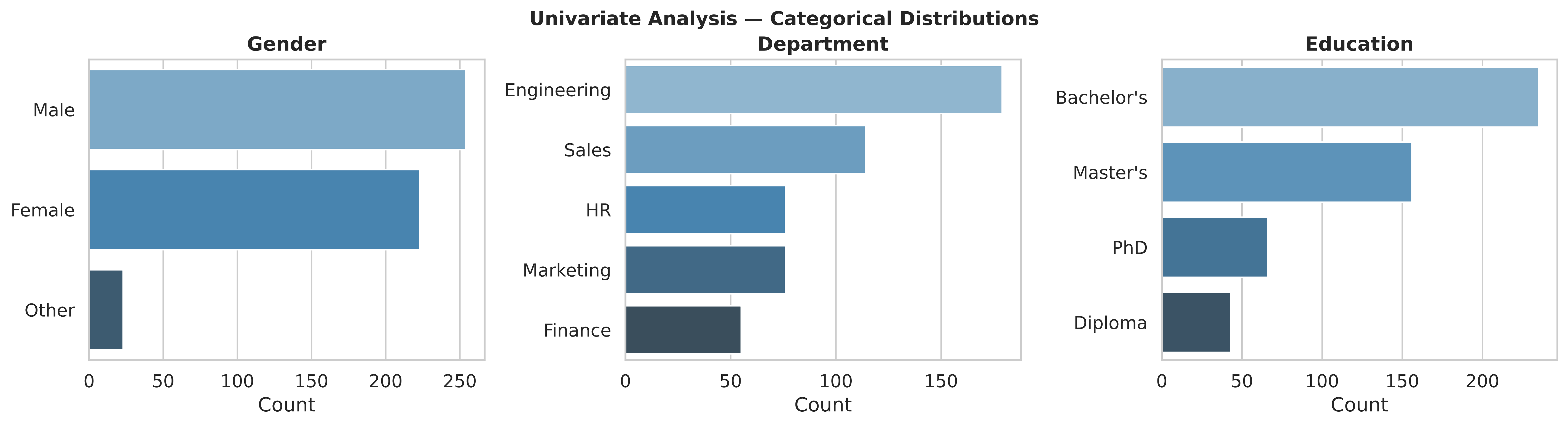
Thiranex Data Science Internship — Task 3

22 June 2026

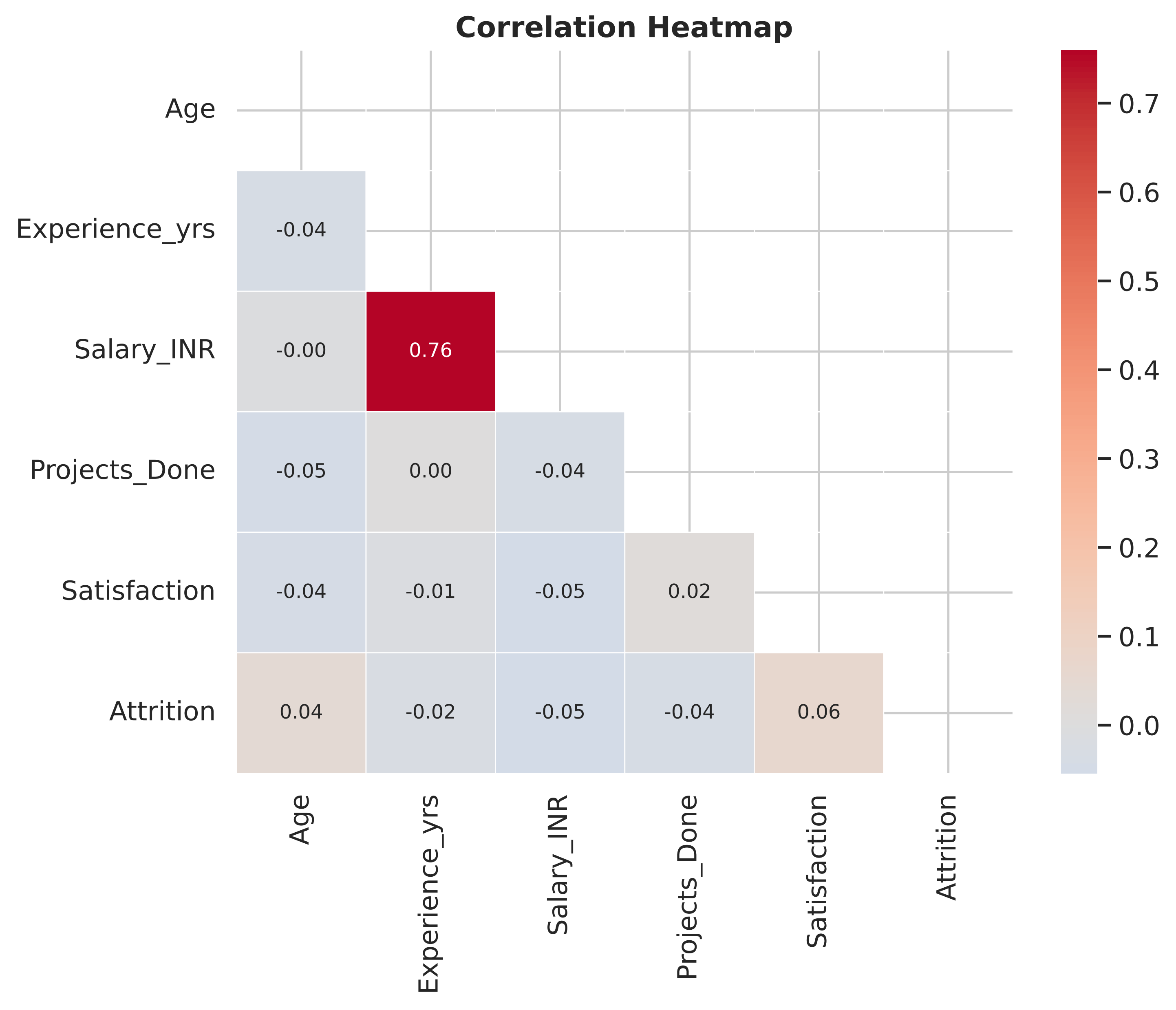
Statistical Summary — Numerical Features

	mean	std	min	50%	max	skewness	kurtosis	cv_%
Age	40.73	10.63	22.0	41.0	59.0	-0.03	-1.09	26.1
Experience_yrs	4.68	4.71	0.0	3.2	30.0	1.93	5.02	100.71
Salary_INR	55396.8	16596.51	20000.0	53750.0	141400.0	1.04	2.47	29.96
Projects_Done	3.96	2.05	0.0	4.0	11.0	0.54	0.12	51.72
Satisfaction	2.95	1.41	1.0	3.0	5.0	0.12	-1.29	47.79
Attrition	0.17	0.38	0.0	0.0	1.0	1.76	1.11	221.18


Univariate Analysis — Numerical Distributions
Age
skew=-0.03
Experience_yrs
skew=1.93
Salary_INR
skew=1.04
Projects_Done
skew=0.54
Satisfaction
skew=0.12
Attrition
skew=1.76
Count

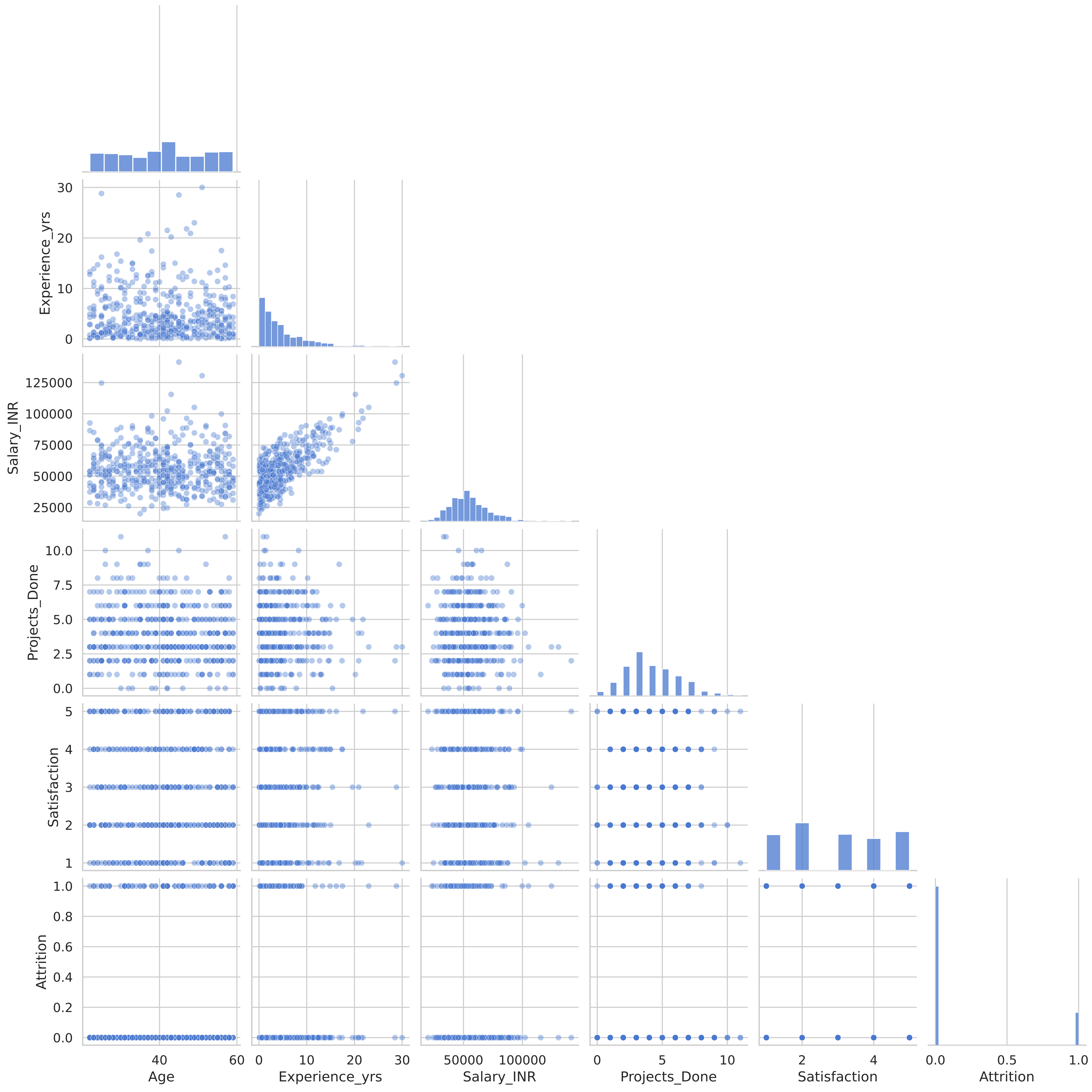


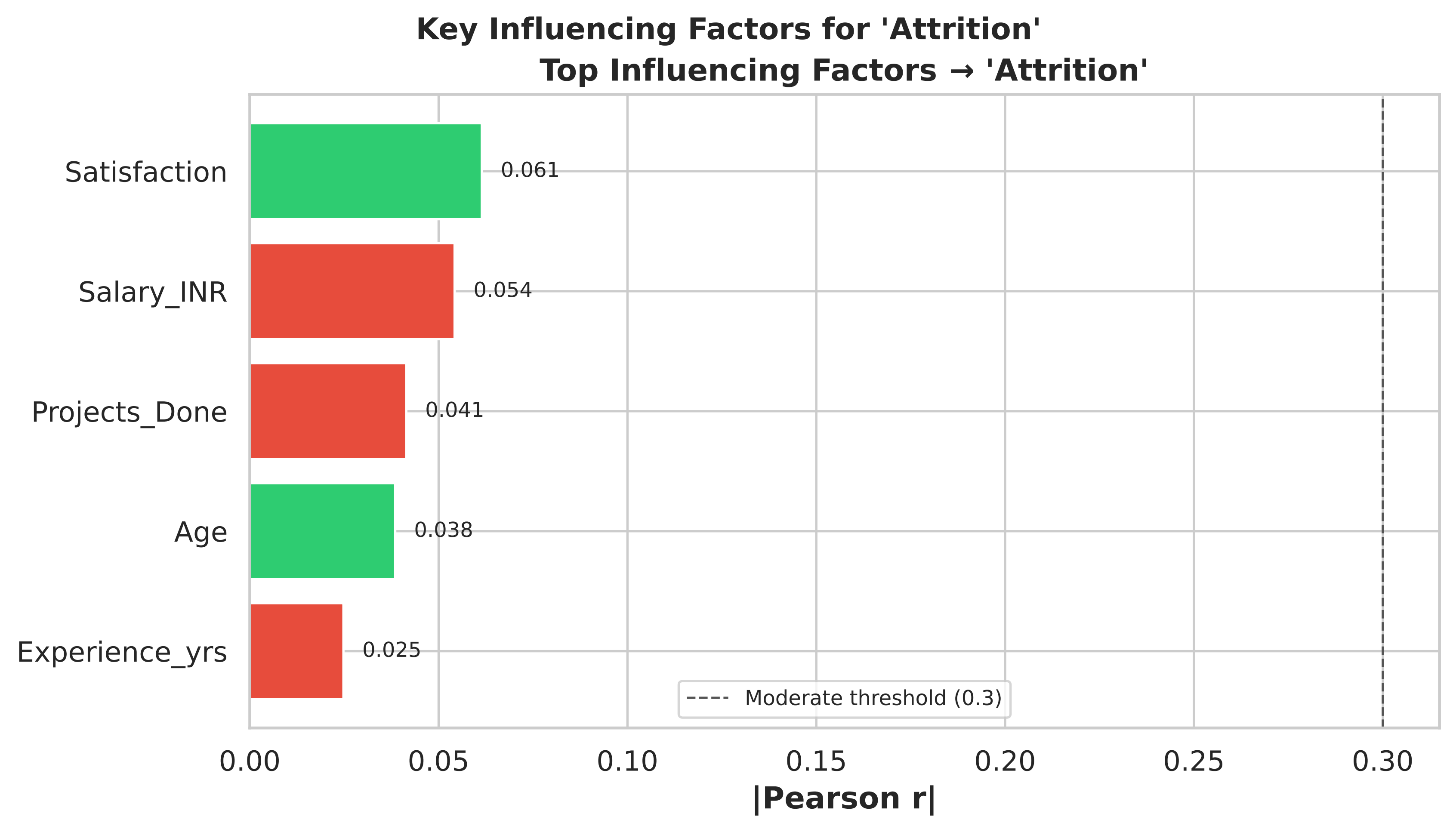
Univariate Analysis — Categorical Distributions
Gender
Male
Female
Other
0
50
100
150
200
250
Count
Department
Engineering
Sales
HR
Marketing
Finance
0
50
100
150
Count
Education
Bachelor's
Master's
PhD
Diploma
0
50
100
150
200
Count

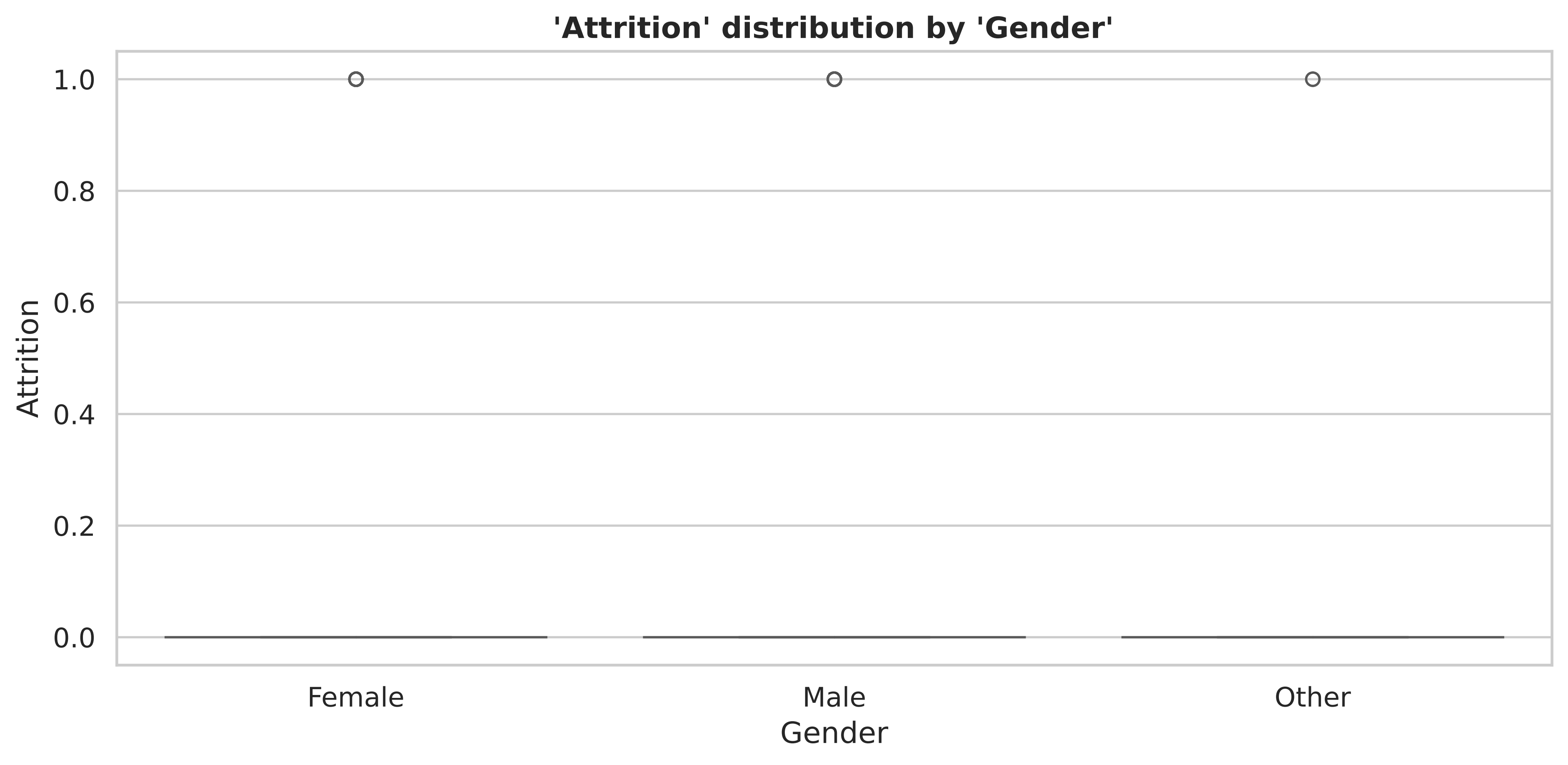


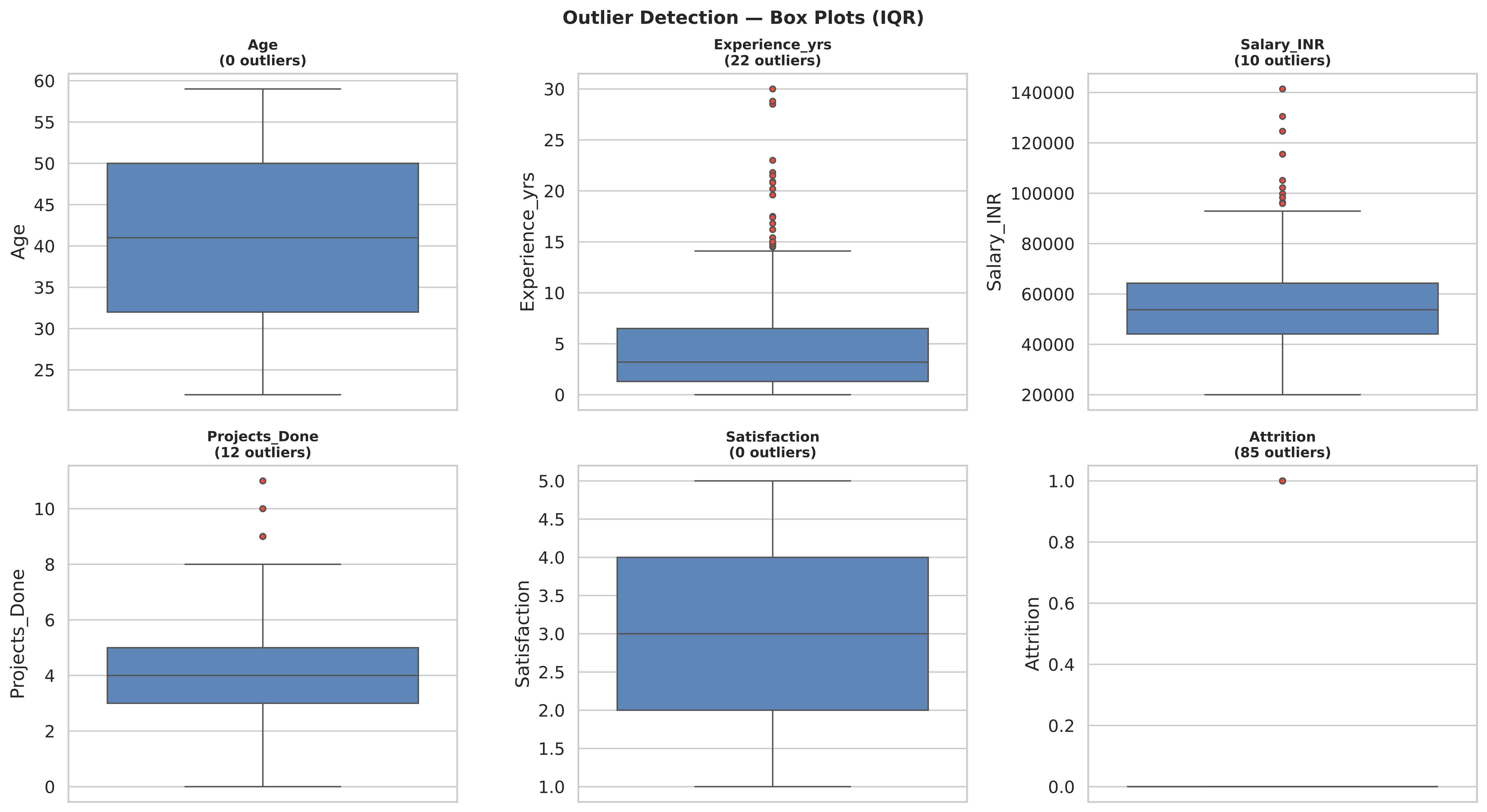
Correlation Heatmap
Age
Experience_yrs
Salary_INR
Projects_Done
Satisfaction
Attrition
-0.04
-0.00
0.76
-0.05
0.00
-0.04
-0.04
-0.01
-0.05
0.02
0.04
-0.02
-0.05
-0.04
0.06
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

Pair Plot — Numerical Features




Key Influencing Factors for 'Attrition'
Top Influencing Factors → 'Attrition'
Satisfaction
Salary_INR
Projects_Done
Age
Experience_yrs
0.061
0.054
0.041
0.038
0.025
Moderate threshold (0.3)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
|Pearson r|


'Attrition' distribution by 'Gender'
1.0
0.8
0.6
0.4
0.2
0.0
Attrition
Female
Male
Other
Gender


Outlier Detection — Box Plots (IQR)
Age
(0 outliers)
Experience_yrs
(22 outliers)
Salary_INR
(10 outliers)
Projects_Done
(12 outliers)
Satisfaction
(0 outliers)
Attrition
(85 outliers)
Age
Experience_yrs
Salary_INR
Projects_Done
Satisfaction
Attrition

Key Insights & Analytical Findings

1. Dataset shape after cleaning: (500, 9) (15 duplicates removed)

2. 'Experience_yrs' is highly right (positive)-skewed (skew=1.93) — consider log-transformation for modelling.

3. 'Salary_INR' is highly right (positive)-skewed (skew=1.04) — consider log-transformation for modelling.

4. 'Attrition' is highly right (positive)-skewed (skew=1.76) — consider log-transformation for modelling.

5. 'Satisfaction' is positively correlated with 'Attrition' (r=0.06), making it a strong predictor.

6. 'Salary_INR' is negatively correlated with 'Attrition' (r=-0.05), making it a strong predictor.